toroid transformer

Bought this on Amazon:

Toroid Core FT240-31 Ferrite



3 inch diameter
0.25 width

Type 77 core material is better, so bought some of them as well, type FT140-77, from Amazon. OD=35.55 mm, ID=23mm.

According to online inductance calculator at https://coil32.net/online-calculators/ferrite-torroid-calculator.html, inductance for N turns of AWG 20 wire is 2 uH X N^2 of inductance.
So 10 turns on the primary would have an inductance of 200 uH.
(Permeability of the type 77 ferrite is 2000).
resonance with 600 pF piezo has omega=1/sqrt(LC) = 1/sqrt(600e-12*200E-6) so f= 400 kHz. Impedance at 100 kHz is omega*L = 2* pi* 1e5*200e-6 = 125 Ohms.
If we have 1:10 ratio of primary to secondary turns, then

From Wikipedia:

Combining Eq. 3 & Eq. 4 with this endnote[b][4] gives the ideal transformer identity:

$$\frac{V_\mathrm{P}}{V_\mathrm{S}} = \frac{I_\mathrm{S}}{I_\mathrm{P}} = \frac{N_\mathrm{P}}{N_\mathrm{S}} = \sqrt{\frac{L_\mathrm{P}}{L_\mathrm{S}}} = a \qquad \textbf{(Eq. 5)}$$

where L is winding self-inductance.

By Ohm's law and ideal transformer identity:

$$Z_\mathrm{L} = \frac{V_\mathrm{S}}{I_\mathrm{S}} \qquad \textbf{(Eq. 6)}$$

$$Z'_\mathrm{L} = \frac{V_\mathrm{P}}{I_\mathrm{P}} = \frac{aV_\mathrm{S}}{I_\mathrm{S}/a} = a^2\frac{V_\mathrm{S}}{I_\mathrm{S}} = a^2 Z_\mathrm{L} \qquad \textbf{(Eq. 7)}$$

where Z_L is the load impedance of the secondary circuit & Z'_L is the apparent load or driving point impedance of the primary circuit, the superscript $'$ denoting referred to the primary.

So if we want to boost applied voltage to piezo from 10V to 100V, we need a turns ratio of 10 turns on secondary for every turn on primary.
Impedance presented to drive circuit then looks 100X higher.

Our fast power amp has output impedance of Z=V/I = 10V/6A = 1.6 Ohms.